

Complexity Through Compression: Hierarchical Predictive Control

Deep Control: from recursive compression to
multiscale reasoning and planning

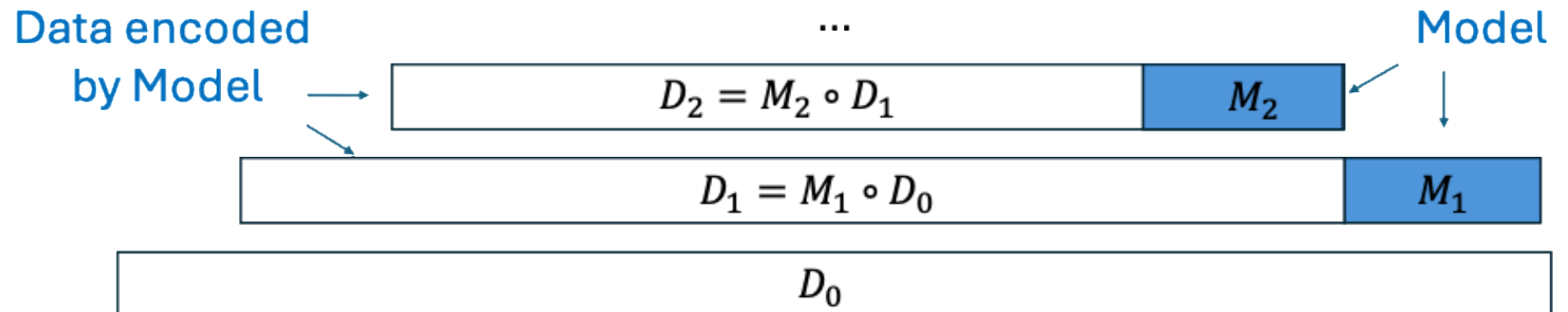
Sergey Shumsky
Chief Scientist, Symbolic Mind Inc.

Compression can create hierarchy

- Good models compress data
- Redundancy exists at multiple scales
- Recursive compression can create structured latent variables
- These variables can support prediction, abstraction, and control

MDL as a learning principle

- Best model minimizes:
 model description length + encoded data length
- Compression identifies reusable structure
- Repeated compression can reveal hierarchy



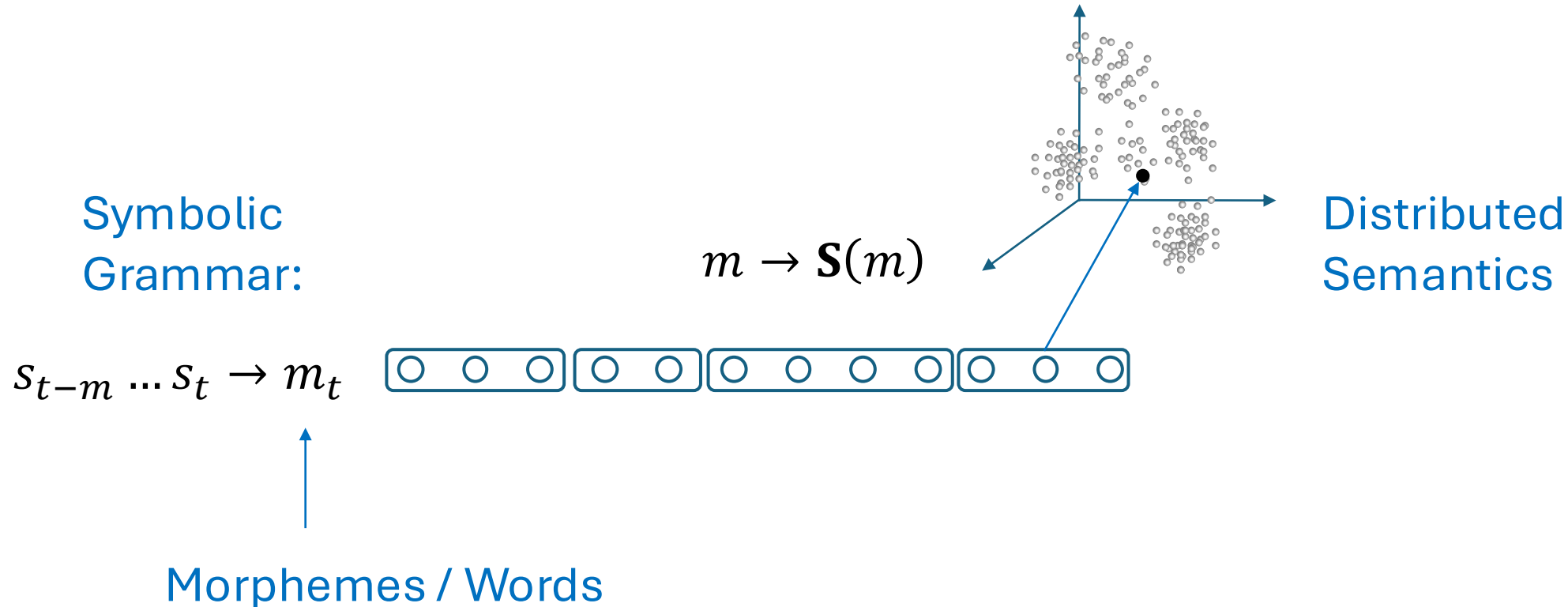
Two forms of compression: structure and meaning

Lossless compression

- Symbolic grammar
- Reusable combinatorial structure

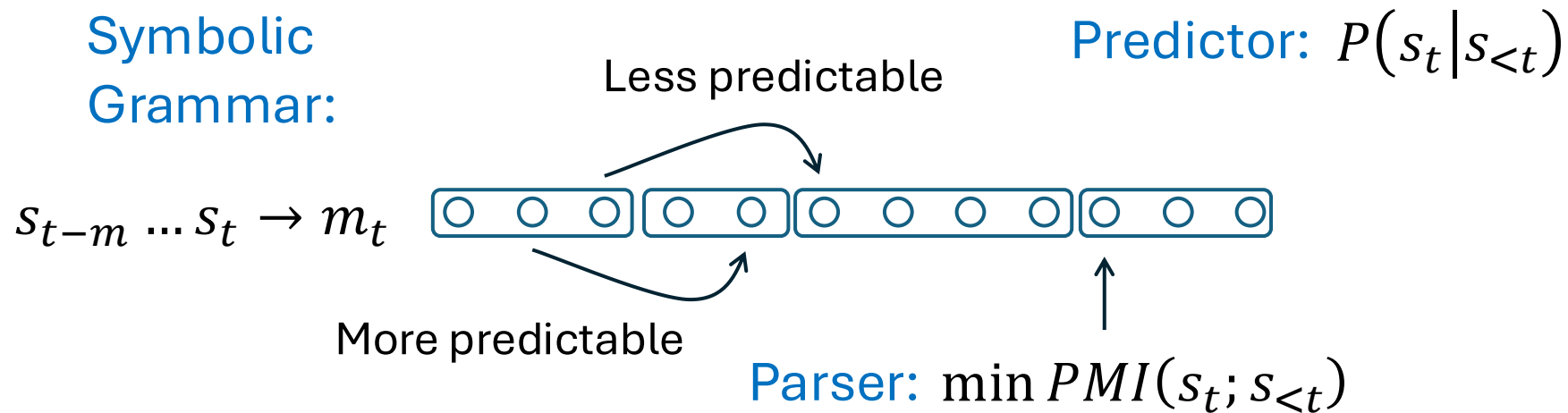
Lossy compression

- Semantic coding
- Compressed context statistics



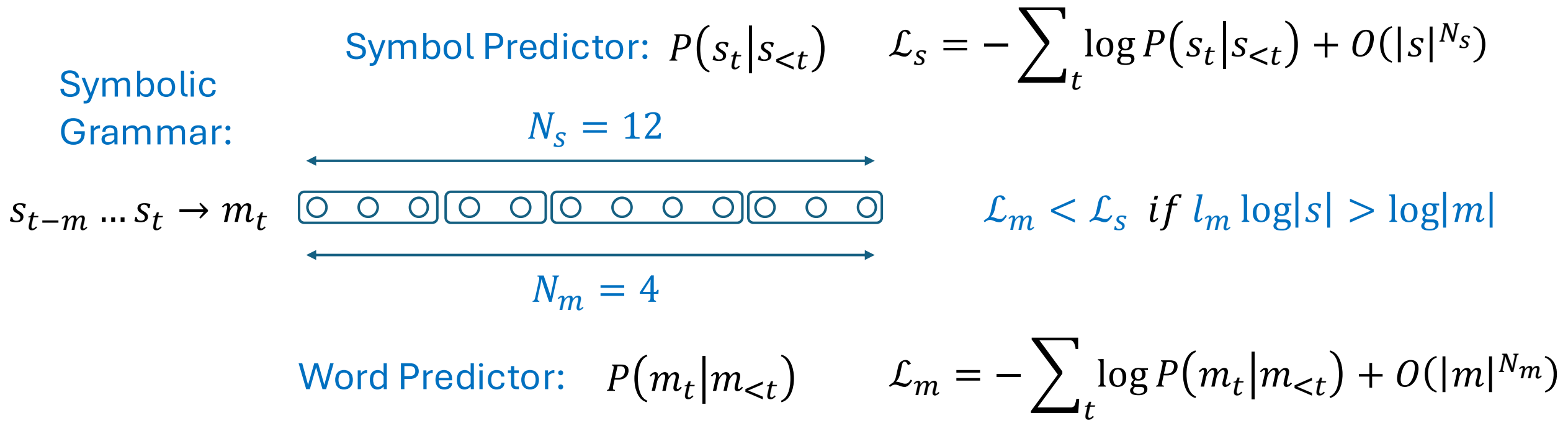
Grammar as compression of a symbol stream

- Learn a symbol predictor
- Detect changes in local predictability
- Parse stream into reusable units
- Build a vocabulary of words or morphemes



Words can shorten description length

- Symbol-level prediction often needs long contexts
- Word-level prediction can use shorter contexts
- If total description length decreases, the new representation is justified

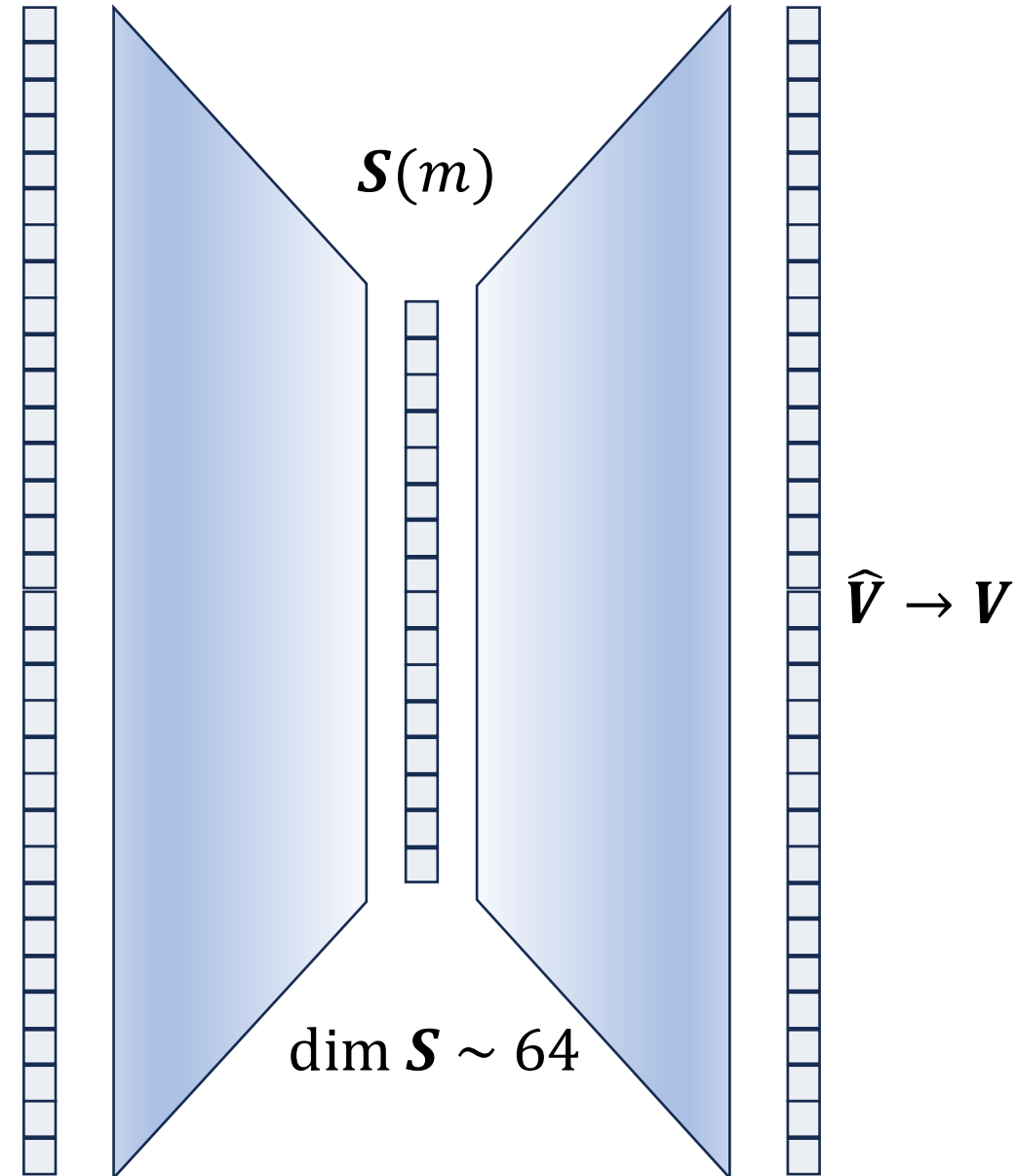


Semantics as compressed context statistics

- Represent each word by its usage context
- Start with high-dimensional distributional vectors
- Compress into short semantic codes
- Preserve neighborhood structure approximately

$$PMI(m, m_i) = V$$

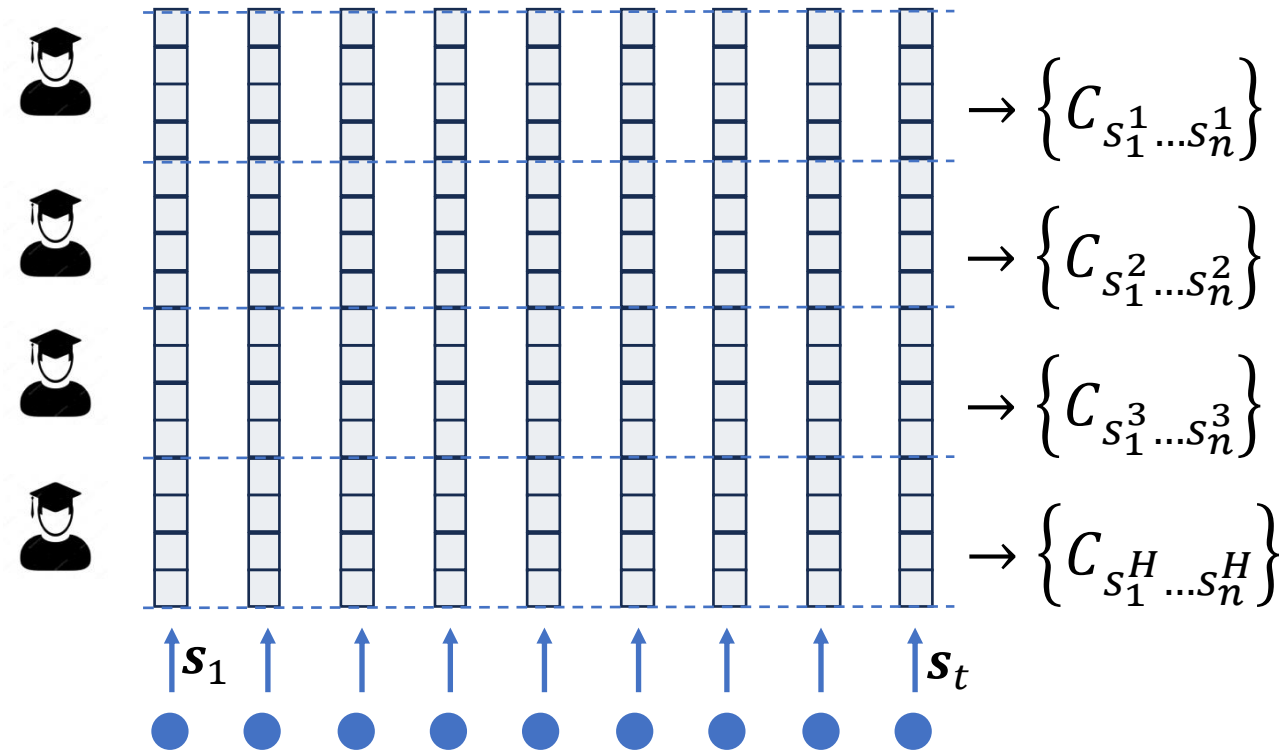
$$\dim V \sim 10^5$$



Multi-head design

- One code is hard to model symbolically
- Several low-bit streams are easier
- Each head captures partial regularities
- Together they act as a committee of experts

Symbolic Heads



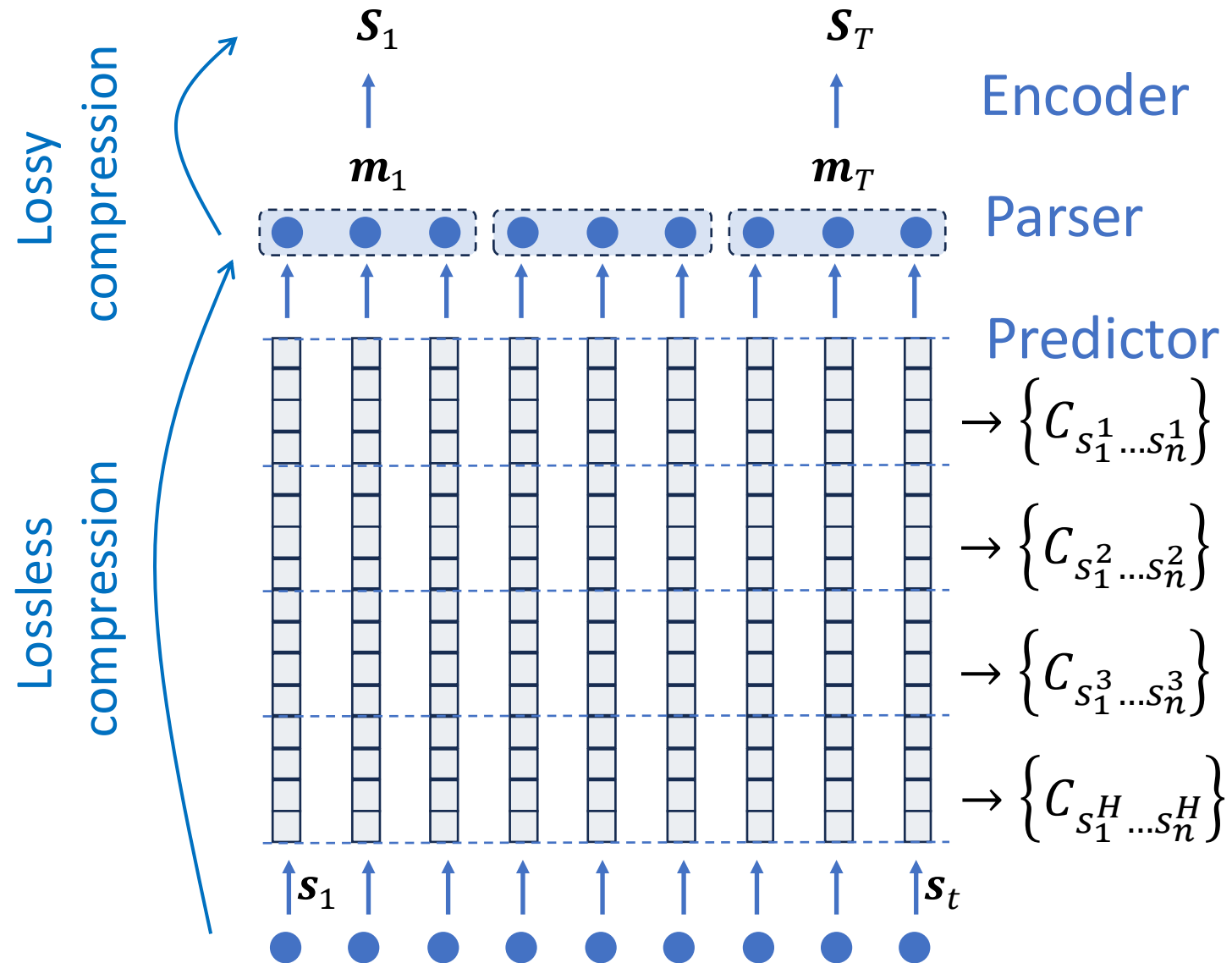
Predictor $P(s_t | s_{<t}) = \prod_h P(s_t^h | s_{<t}^h)$

Recursive stack of languages

Layer L :

- Parse symbolic stream
- Learn vocabulary
- Encode learned words into higher-level symbols

Repeat at layer $L + 1$

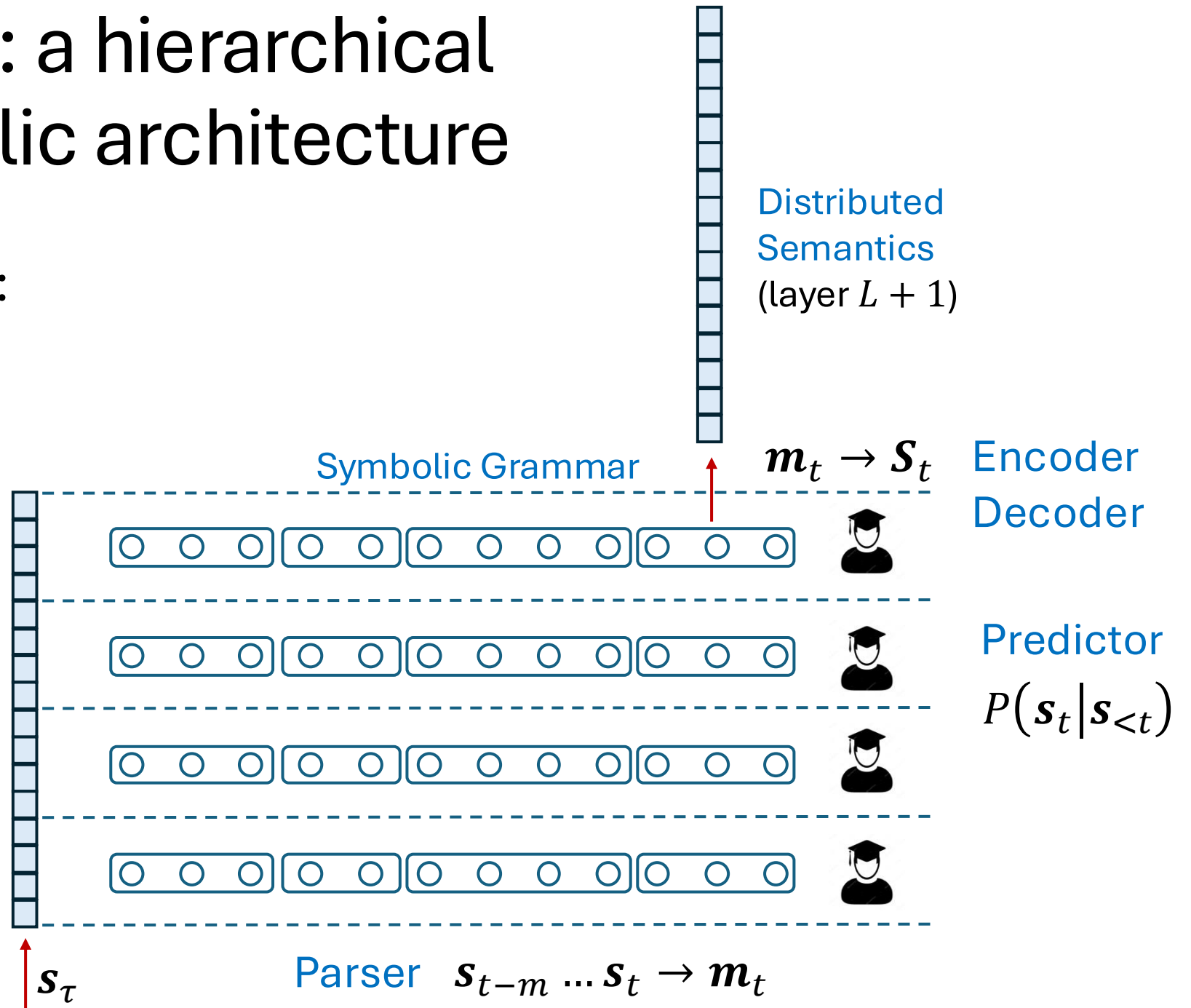


Deep Control: a hierarchical neuro-symbolic architecture

Each layer contains:

- Predictor
- Parser
- Encoder
- Decoder

Distributed
Semantics
(layer L)



Inference as compression-driven search

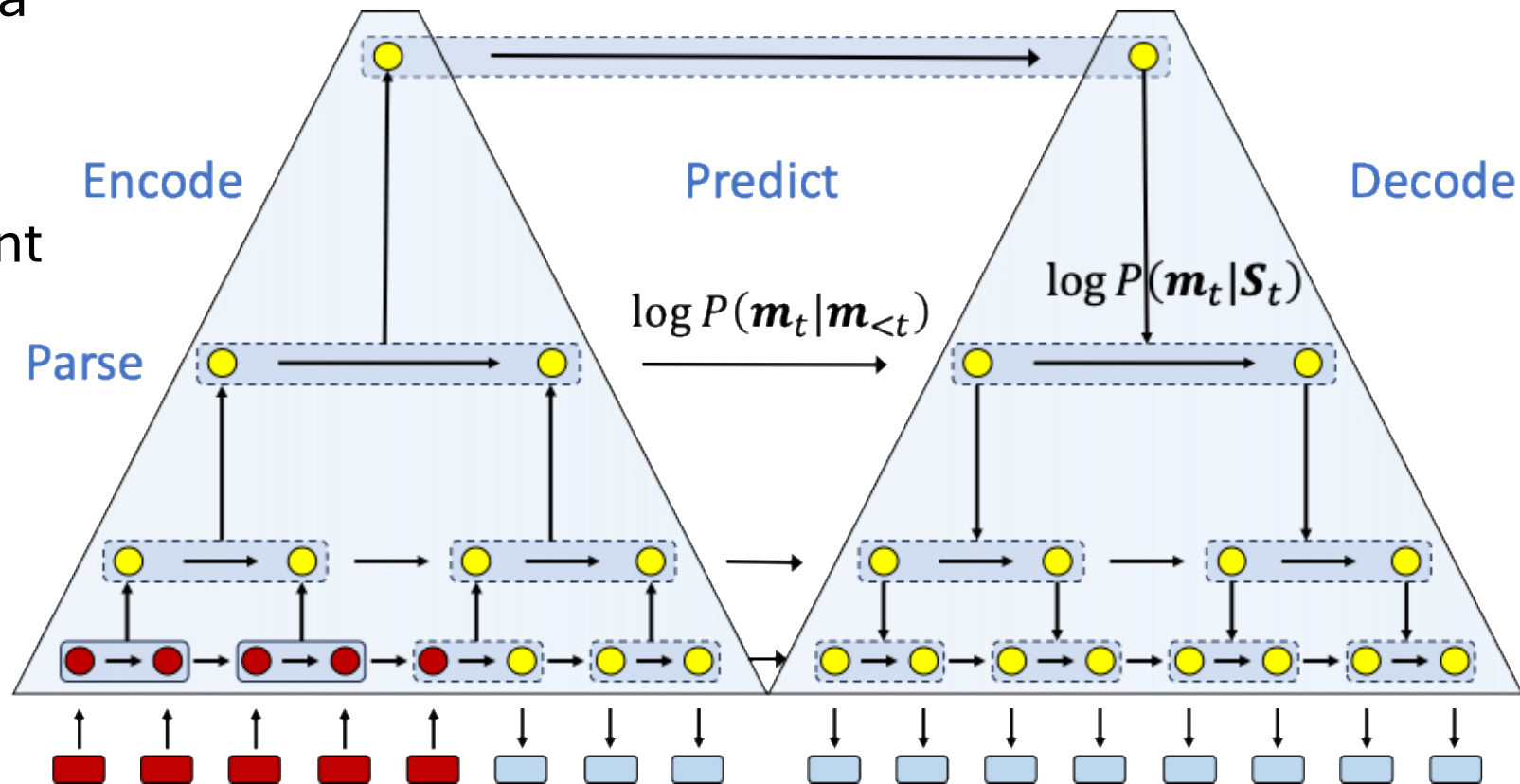
Learning:

- choose the model that best compresses data

$$\mathcal{L} \approx - \sum_t [\log P(\mathbf{m}_t | \mathbf{m}_{<t}) + \log P(\mathbf{m}_t | \mathbf{S}_t)] \rightarrow \min$$

Inference:

- search for the most compact and coherent representation under the model



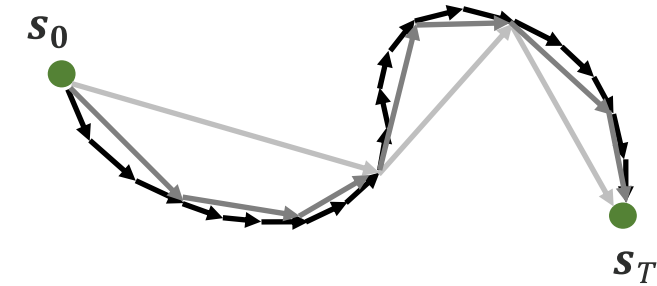
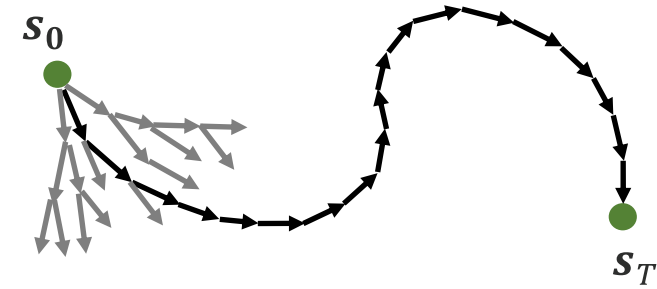
From local prediction to structured reasoning

Transformers:

- strong local next-token prediction

Deep Control:

- multiscale internal structure
- explicit intermediate abstractions
- search for compatible subgoals and descriptions



Current implementation and early evidence

Current status:

- **Eva** is a working prototype of a compact language model
- Learning pipeline scales linearly with processed data
- Current training speed ~ **1GB/hour** on one CPU server
- This makes **terabyte-scale training** feasible on modest CPU hardware
- On BLiMP, performance is comparable to transformer baseline trained on the same data

Next step:

- Extend the architecture toward a hierarchical reasoning engine for agents and robots

What higher layers may represent

Reasoning ~ 3-5 arguments

Arguments ~ 3-5 concepts

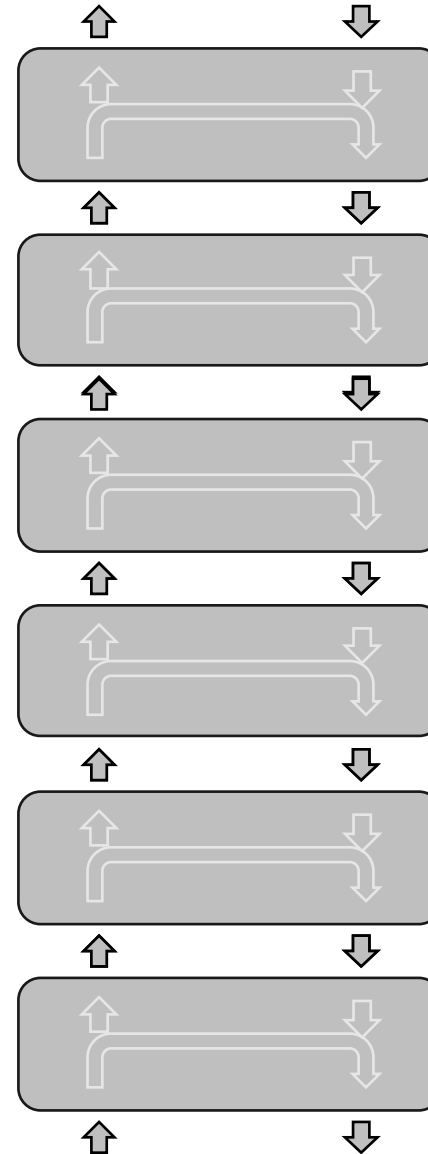
Concepts ~ 3-5 sentences

Sentences ~ 3-5 phrases

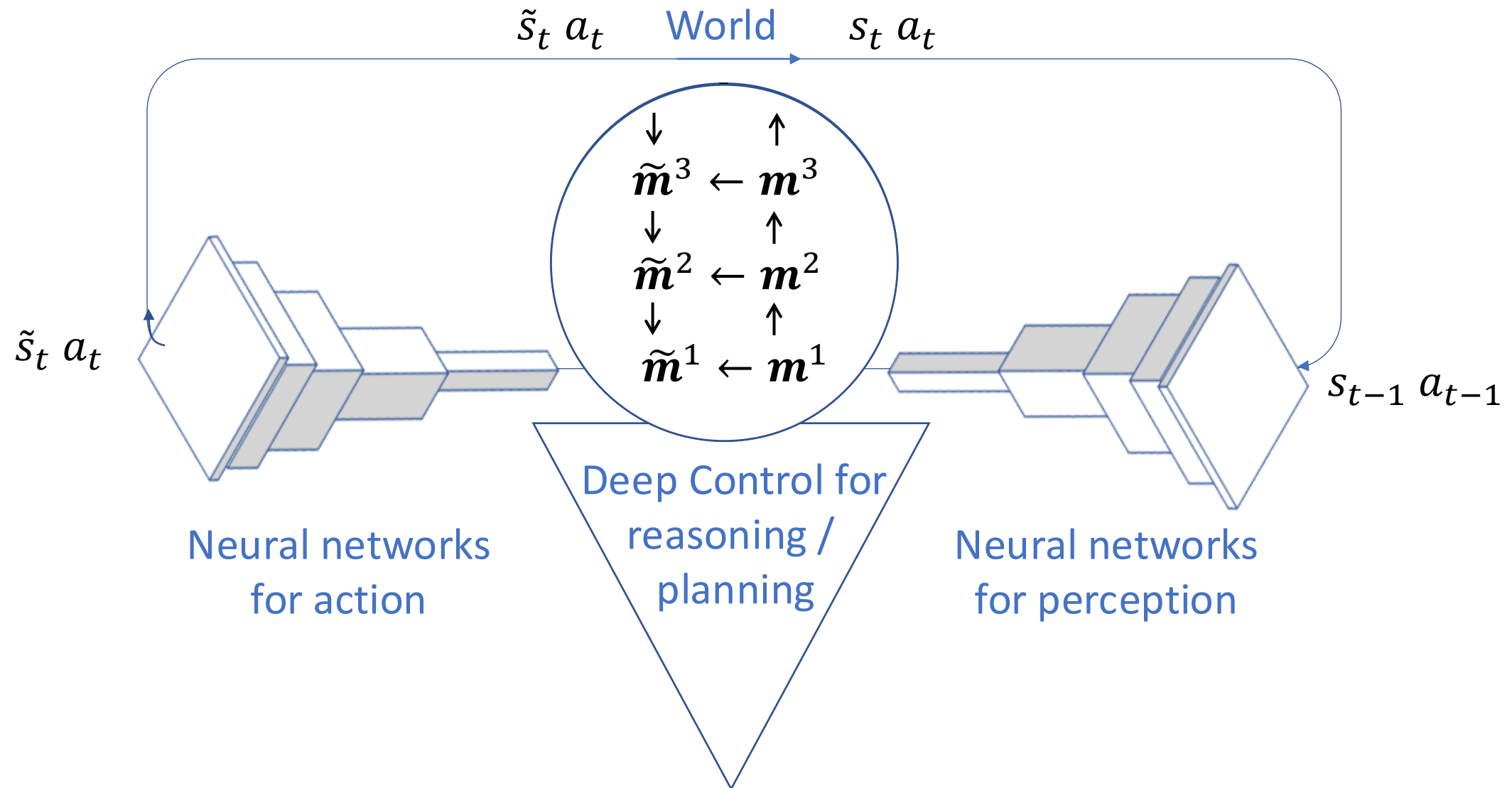
Phrases ~ 3-5 words

Words ~ 3-5 characters

Encode → Reason



Toward a reasoning layer for embodied agents



Takeaways

- Recursive compression can create hierarchy
- Hierarchy can support abstraction, prediction, and control
- Deep Control is an architectural attempt to operationalize this principle